

Literal Residual Grouping and the Native Gram: Divisor-Collision Parametrization, Exact Coarea Singular Values, and a Provenance Intertwining

Liang Wang

School of Artificial Intelligence and Automation

Huazhong University of Science and Technology, Wuhan 430074, P.R. China

wangliang.f@gmail.com

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Abstract

The current TPC physical route contains two distinct identifications. Source records first collide at the literal residual label

$$q(u) = (\gamma_u, j_u, s_u), \quad s_u = d(m_u)r_u,$$

and residual labels may then collide after passage to a native frequency label $\pi(q(u))$. Treating these operations as one informal “grouping” hides both their kernels and their losses.

We separate them as a weighted residual coarea map

$$\mathcal{R}_a e_u = a_u e_{q(u)}$$

and an unweighted native forgetting map $\mathcal{P}e_y = e_{\pi(y)}$. Three exact results follow. First, $dr = d'r'$ has the complete coprime parametrization

$$d = ga, \quad d' = gb, \quad r = bt, \quad r' = at, \quad (a, b) = 1.$$

Second,

$$\mathcal{R}_a \mathcal{R}_a^* = \text{diag}(W_y), \quad W_y = \sum_{q(u)=y} |a_u|^2,$$

so every nonzero singular value of $\mathcal{R}_a$ is exactly $\sqrt{W_y}$. Third, the residual and native Grams are

$$G_{\text{res}} = \mathcal{R}_a^* \mathcal{R}_a, \quad G_{\text{nat}} = \mathcal{R}_a^* \mathcal{P}^* \mathcal{P} \mathcal{R}_a,$$

and their difference is exactly the exchange between unequal residuals sharing one native label. For any upstream synthesis $\mathcal{S}$, the complete native observation satisfies

$$(\mathcal{P} \mathcal{R}_a \mathcal{S})^* (\mathcal{P} \mathcal{R}_a \mathcal{S}) = \mathcal{S}^* G_{\text{nat}} \mathcal{S}.$$

The upper ℓ^2 -energy factor is the largest native multiplicity, and the corresponding ℓ^2 -amplitude norm is its square root. Neither is automatically the H9 max-packet-to-scalar cost. When a native fiber contains at least two residual labels, no positive lower transfer exists on the full space without excluding the cancellation kernel: two residuals may add or cancel exactly after native forgetting. Thus this paper supplies an exact L0 algebra and an L1 provenance dictionary, but no growing multiplicity bound, fixed- h_0 L2 estimate, parity breakthrough, or prime-pair theorem.

Keywords: divisor collision; residual coarea; native Gram; alias exchange; physical provenance; fixed shift.

Literal convention. Every record u , coefficient a_u , divisor value $d(m_u)$, residual coordinate r_u , and surviving label is inherited. Neither a coefficient nor a collision multiplicity may be normalized away after the fact. All Hilbert spaces below use their declared counting measures.

1 Why two groupings are necessary

TPC-46 proves that Bohr–Fourier orthogonality retains a literal residual label and that positivity-only localization can spend the full alias mass [3]. TPC-58 introduces a native affine alias Gram, while TPC-61 identifies the remaining cofactor/parity kernel [4, 5]. TPC-113 gives the correct quotient condition number for any declared finite physical synthesis [2]. The missing dictionary is the exact map between the source, residual, and native levels.

Let U be a finite set of source records. A record carries

$$u = (m_u, r_u, \gamma_u, j_u, \xi_u), \quad a_u \in \mathbb{C}, \quad d_u = d(m_u) \in \mathbb{N}.$$

The auxiliary label ξ_u records any remaining provenance but does not enter the residual key

$$q(u) = (\gamma_u, j_u, d_u r_u) \in Y.$$

Let $\pi : Y \rightarrow I$ be the exact native label map supplied by the application. We do not assume it is injective.

2 The divisor-collision normal form

Lemma 2.1 (Complete product-collision parametrization). *For positive integers d, d', r, r' , the equality $dr = d'r'$ holds if and only if there exist $g, a, b, t \in \mathbb{N}$ such that*

$$d = ga, \quad d' = gb, \quad (a, b) = 1, \quad r = bt, \quad r' = at.$$

The parameters are unique with $g = (d, d')$.

Proof. Write $d = ga$, $d' = gb$, where $g = (d, d')$ and $(a, b) = 1$. Then $dr = d'r'$ is $ar = br'$. Coprimality gives $b \mid r$ and $a \mid r'$, hence $r = bt$ and $r' = at$ for the same t . The converse is immediate, and the gcd choice proves uniqueness. $\square$

Remark 2.2 (What the lemma does not count). [Lemma 2.1](#) identifies every arithmetic collision, but it does not bound how many admissible quadruples occur in the actual active support. Such a bound must retain the masks on m, r, γ and j . Replacing them by a rectangular ambient box would change the physical fiber.

3 Weighted residual coarea

Let $\{e_u : u \in U\}$ and $\{e_y : y \in Y\}$ be orthonormal bases. Define

$$\mathcal{R}_a : \mathbb{C}^U \rightarrow \mathbb{C}^Y, \quad \mathcal{R}_a e_u = a_u e_{q(u)}.$$

Zero-weight records are allowed. Put

$$W_y = \sum_{\substack{u \in U \\ q(u)=y}} |a_u|^2.$$

Theorem 3.1 (Exact coarea singular values). *The weighted grouping map satisfies*

$$\boxed{\mathcal{R}_a \mathcal{R}_a^* = \text{diag}_{y \in Y}(W_y)}.$$

Consequently its nonzero singular values, with multiplicity, are $\{\sqrt{W_y} : W_y > 0\}$, and

$$\begin{aligned} \|\mathcal{R}_a\|^2 &= \max_y W_y, & \|\mathcal{R}_a^\dagger\|^2 &= \frac{1}{\min_{W_y > 0} W_y}, \\ \kappa_{\text{grp}}(\mathcal{R}_a)^2 &= \frac{\max_y W_y}{\min_{W_y > 0} W_y}. \end{aligned}$$

Proof. For $y, y' \in Y$,

$$\langle e_y, \mathcal{R}_a \mathcal{R}_a^* e_{y'} \rangle = \sum_{u \in U} |a_u|^2 \mathbf{1}_{\{q(u)=y\}} \mathbf{1}_{\{q(u)=y'\}}.$$

The expression vanishes for $y \neq y'$ and equals W_y on the diagonal. The remaining assertions follow from the singular-value decomposition and the quotient convention of TPC-113. $\square$

Remark 3.2 (Weighted, not cardinality-only). If $|a_u|$ varies, the fiber cardinality is not the coarea weight. The exact quantity is W_y . A divisor-bound estimate for the number of records is therefore insufficient unless the physical weights are controlled on the same support.

4 The native Gram and alias exchange

Define the native forgetting map

$$\mathcal{P} : \mathbb{C}^Y \rightarrow \mathbb{C}^I, \quad \mathcal{P} e_y = e_{\pi(y)}.$$

For $i \in I$, write

$$k_i = \#\{y \in Y : \pi(y) = i\}.$$

Proposition 4.1 (Native multiplicity spectrum). *One has*

$$\mathcal{P} \mathcal{P}^* = \text{diag}_{i \in I}(k_i), \quad \|\mathcal{P}\|^2 = \max_i k_i.$$

The nonzero singular values are $\sqrt{k_i}$ for $k_i > 0$.

Proof. The rows of $\mathcal{P}$ have disjoint supports, and row i has exactly k_i unit entries. The result is [theorem 3.1](#) with all weights equal to one and grouping map π . $\square$

The two source-side Grams are

$$G_{\text{res}} = \mathcal{R}_a^* \mathcal{R}_a, \quad G_{\text{nat}} = \mathcal{R}_a^* \mathcal{P}^* \mathcal{P} \mathcal{R}_a.$$

Theorem 4.2 (Exact Gram dictionary). *For all $u, v \in U$,*

$$\begin{aligned} (G_{\text{res}})_{uv} &= \overline{a_u} a_v \mathbf{1}_{\{q(u)=q(v)\}}, \\ (G_{\text{nat}})_{uv} &= \overline{a_u} a_v \mathbf{1}_{\{\pi(q(u))=\pi(q(v))\}}, \end{aligned}$$

and hence

$$(G_{\text{nat}} - G_{\text{res}})_{uv} = \overline{a_u} a_v \mathbf{1}_{\{q(u) \neq q(v), \pi(q(u))=\pi(q(v))\}}.$$

Thus the difference contains exactly the exchange between distinct residual fibers that alias to the same native label.

Proof. The first formula is the matrix coefficient of $\mathcal{R}_a^* \mathcal{R}_a$. Since $(\mathcal{P}^* \mathcal{P})_{yy'} = \mathbf{1}_{\{\pi(y)=\pi(y')\}}$, insertion between $\mathcal{R}_a^*$ and $\mathcal{R}_a$ gives the second. Subtraction leaves precisely equal-native, unequal-residual pairs. $\square$

Remark 4.3 (The exchange is not a positive correction). Although both G_{nat} and G_{res} are positive semidefinite, their difference need not be. The exchange terms can add or cancel depending on the coefficient phases, so they may not be entered as an automatic energy gain.

Corollary 4.4 (Exact provenance intertwining). *Let $\mathcal{S} : \mathbb{C}^\Theta \rightarrow \mathbb{C}^U$ be any upstream literal synthesis and set $T = \mathcal{P} \mathcal{R}_a \mathcal{S}$. Then*

$$T^* T = \mathcal{S}^* G_{\text{nat}} \mathcal{S}.$$

In particular, every native quadratic estimate on the inherited coefficient is a statement about this compressed physical Gram, not about G_{res} alone.

Proof. Expand $T^* T$ and use the definition of G_{nat} . $\square$

5 Sharp lower-transfer obstruction

Proposition 5.1 (Addition and cancellation in one native fiber). *Suppose $y_1 \neq y_2$ and $\pi(y_1) = \pi(y_2)$. On their span,*

$$\|e_{y_1} + e_{y_2}\|^2 = 2, \quad \|\mathcal{P}(e_{y_1} + e_{y_2})\|^2 = 4,$$

whereas

$$\|e_{y_1} - e_{y_2}\|^2 = 2, \quad \mathcal{P}(e_{y_1} - e_{y_2}) = 0.$$

Therefore the upper energy factor $k_{\max}$ is sharp, while no positive lower comparison $\|\mathcal{P}z\|^2 \geq c\|z\|^2$ can hold on the full residual space.

Proof. Both residual basis vectors map to the same native basis vector. The displayed identities follow by addition and subtraction. $\square$

Remark 5.2 (Collision is necessary for this obstruction). If every native fiber is a singleton, then $\mathcal{P}$ is an isometry onto its range and the cancellation example is unavailable. An actual stop certificate must also show that the distinguished coefficient, or the relevant range of $\mathcal{R}_a\mathcal{S}$, meets the collision kernel.

Remark 5.3 (Why separate losses must not be multiplied blindly). [Corollary 4.4](#) computes the complete Gram before any condition-number estimate. Bounding $\mathcal{S}$, $\mathcal{R}_a$, and $\mathcal{P}$ separately can be convenient, but multiplying the three worst-case costs may charge directions that never coexist. The native physical condition number should be measured from $\mathcal{S}^*G_{\text{nat}}\mathcal{S}$ whenever that matrix is available.

6 Finite exact certificate

The script `experiments/tpc114_grouping_certificate.py` uses rational arithmetic. It enumerates product collisions, constructs $\mathcal{R}_a, \mathcal{P}, G_{\text{res}}, G_{\text{nat}}$, and checks:

1. the coprime normal form in [lemma 2.1](#);
2. $\mathcal{R}_a\mathcal{R}_a^* = \text{diag}(W_y)$;
3. $\mathcal{P}\mathcal{P}^* = \text{diag}(k_i)$;
4. both entrywise Gram formulas and the exchange identity; and
5. the sharp addition/cancellation pair.

The JSON certificate is finite exact algebra. It is not evidence that actual TPC fibers have bounded multiplicity.

7 Stop/go verdict and claim boundary

Status	Required next certificate	Consequence
Go	Actual weighted fibers W_y , native degrees k_i , and the relevant positive spectrum of $\mathcal{S}^*G_{\text{nat}}\mathcal{S}$ are controlled on the inherited support	Charge the proved physical loss once and attempt fixed- h_0 localization.
Stop this projection	The actual coefficient has a nonnegligible component in $\ker(\mathcal{P}\mathcal{R}_a\mathcal{S})$, or the strict end-point is exhausted	Native forgetting cannot recover the required direction; retain more labels or change the observation.
Open	Only divisor counts, ambient supports, or unweighted fibers are known	Do not infer a native frame bound.

Established. [Lemma 2.1](#), [theorems 3.1](#) and [4.2](#), [propositions 4.1](#) and [5.1](#), and [corollary 4.4](#) give a complete finite-dimensional dictionary. The matrix identities and sharp obstruction are L0. Declaring $q(u) = (\gamma, j, d(m)r)$ and retaining all literal weights is the L1 arithmetic interface anticipated by TPC-MVP1 [\[1\]](#).

Not established. We prove no asymptotic bound for W_y , k_i , or the positive spectrum of the actual growing native Gram. We do not identify a protected complement to the cancellation kernel, recover one prescribed h_0 , or prove a fixed-power arithmetic saving. There is no automatic conversion from the ℓ^2 norm $\sqrt{k_{\max}}$ to the H9 max-packet-to-scalar norm; that requires the actual operator chain. There is no L2 advance, parity breakthrough, Hardy–Littlewood asymptotic, prime-pair lower bound, or twin-prime theorem. Nothing is specialized to $h_0 = 2$.

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